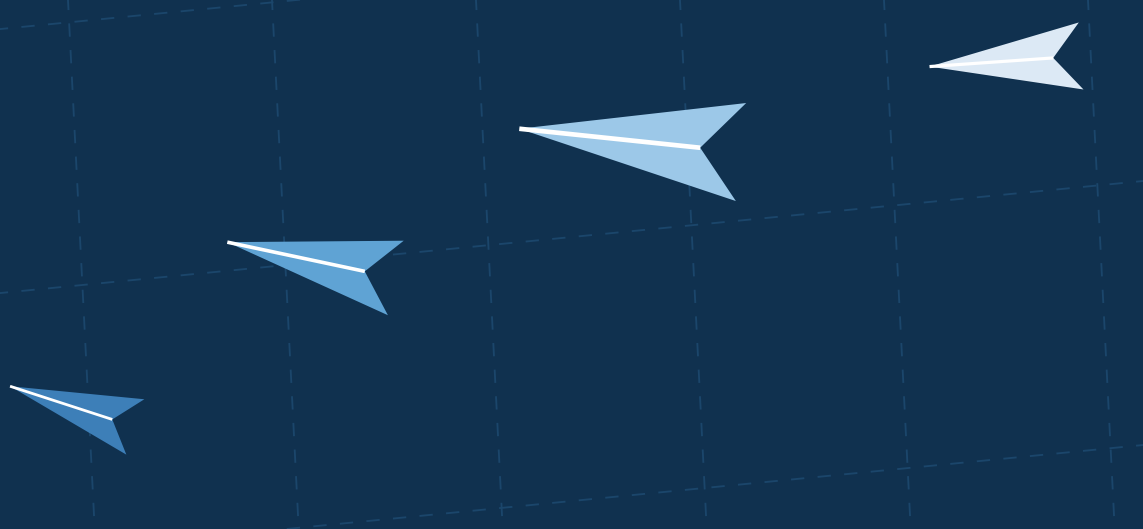




A FIELD MANUAL OF FLYING MACHINES

THE PAPER HANGAR

25 paper airplanes, from the two-minute Classic Dart to a world-record-style glider — with step-by-step fold diagrams, flight ratings, throwing technique and tuning advice for every single design.

5 SQUADRONS

Darts & Speed · Distance Champions · Gliders & Floaters · Stunt & Acrobatic · Exotic & Unusual

Everything folds from a single sheet of letter or A4 paper. No glue. Scissors twice, at most.

Before You Fold

Reading the diagrams

-  Valley fold — fold the paper toward you, along this line. The most common fold.
-  Mountain fold — fold the paper away from you, behind the sheet.
-  Existing crease — a fold you already made, shown for reference.
-  Cut line — the rare design that wants scissors.
-  Movement arrow — which part of the paper moves, and where it goes.

Choosing paper

Standard 20 lb (75–80 gsm) printer paper, letter or A4, is the default for every design in this manual and the right answer 90% of the time. Lighter paper floats better (gliders, the Tumblewing); heavier 24–28 lb paper holds creases harder and fights wind (the Hammer, outdoor darts). Avoid glossy magazine paper — it won't hold a crease — and never fight curled paper: start with a flat sheet.

The golden rules of folding

- Crease like you mean it. Run a fingernail or ruler edge along every fold. Soft creases make soft flyers.
- Symmetry beats everything. The left and right halves must mirror each other exactly; a 2 mm error in a wing fold is the difference between a cruise and a corkscrew.
- Fold on a hard, flat surface. Your lap is where good planes go to die.
- Follow the order. Fold sequences build on earlier creases; skipping ahead breaks the geometry.
- Refold rather than repair. If a fold lands crooked, unfold and redo it — paper forgives one retry.

Why planes fly (the 60-second version)

Every flight is a negotiation between four forces: your throw provides thrust, the wings turn moving air into lift, the paper's weight pulls down, and drag bleeds off speed. Darts minimize drag and rely on thrust; gliders maximize lift and sip their speed slowly. The single most important variable you control is balance: weight concentrated toward the nose keeps the plane pointed where it's going. That's why so many folds in this book exist purely to pack paper into the front.

Throwing & Tuning

The grip

Pinch the fuselage between thumb and forefinger near the plane's balance point — usually a third of the way back from the nose, where it balances on a fingertip. Gripping the tail makes the nose flop; gripping the very nose crushes your ballast.

The release

More flights are ruined by the release than the fold. The plane must leave your hand level, with the wings flat and no sideways wrist-flick. Follow through straight, like throwing a dart at a board.

Match power to plane

Darts want a firm, flat, shoulder-height throw. Distance planes want full power at 30–45° upward. Gliders want either a gentle level launch or a full-strength vertical heave (then they recover at the top and float down). Stunt planes each specify their own — check the spec sheet.

Trim small, trim often

Tuning means bending, not refolding. Plane dives? Bend the back edges of both wings up 2–3 mm ('up-elevator'). Plane climbs then plummets (a stall)? Reduce that bend or throw softer. Veers left? Bend the back of the LEFT wing down slightly, or check that the left winglet isn't leaning. Corkscrews? Wings are uneven — re-level them so both make the same slight 'Y' with the body. Adjust one thing per throw; millimeters are plenty.

The pre-flight check

Nose-on, your plane should look like a shallow letter Y: fuselage vertical, wings level or raised a touch (dihedral), winglets vertical. Squint down the fuselage from behind to spot twists. Ten seconds of checking saves ten throws of confusion.

Flying with respect

Never throw any plane — even paper — at a person's face; a fast dart has a genuinely sharp point. Indoors beats outdoors for everything except the Hammer and the Condor; even a light breeze overwhelms most designs. And pick your planes up: a hangar with a clean floor is a hangar you're allowed back into.

The Hangar

DARTS & SPEED

Classic Dart	EASY	The one every pilot learns first — fast, straight and true.	5
The Arrow	EASY	A dart with winglets — straighter than straight.	6
Bulldog Dart	EASY	A blunt-nosed brawler that survives every crash landing.	8
The Concorde	MEDIUM	Supersonic looks, hallway-dogfight handling.	10
The Sprinter	MEDIUM	A strip-loaded nose for drag-race acceleration.	12

DISTANCE CHAMPIONS

Nakamura Lock	EASY	The locked-nose legend — the best easy plane ever folded.	14
The Suzanne	HARD	Folded in the style of the world-record distance glider.	16
The Hammer	HARD	A dense, armored nose for maximum outdoor range.	18
The Eagle	MEDIUM	A Toda-style competition dart, tuned for clean release.	20
Alpha Strike	MEDIUM	A locked nose plus tall winglets — far and forgiving.	22

GLIDERS & FLOATERS

Stealth Glider	MEDIUM	The hang-time champion — throw it high and count the seconds.	24
Sea Glider	MEDIUM	A moth-winged delta that drifts straight and far.	26
The Albatross	EASY	The all-purpose floater — calm, stable, dependable.	28
The Moth	EASY	Three minutes to fold, all wing, pure float.	30
Sky King	HARD	A duration specialist in the spirit of the 29-second record.	31
The Condor	MEDIUM	A big-winged soarer that loves a high launch point.	33

STUNT & ACROBATIC

The Stunt Plane	MEDIUM	Loops on demand — bend the elevators and let it rip.	35
The Boomerang	HARD	Throw it away and it comes home.	37
The Spinner	MEDIUM	A corkscrewing dart that drills through the air.	39
The Loop	EASY	The easiest trick plane — a guaranteed vertical loop.	41

EXOTIC & UNUSUAL

The Ring Wing	EASY	No wings, no fuselage — a flying hoop that shouldn't work.	43
The Tumblewing	EASY	A walkalong glider you can surf on a sheet of cardboard.	44
The Canard	HARD	A fighter-jet layout with little wings up front.	45
The Hurricane	MEDIUM	A swept flying wing with no body at all.	47
The Dragonfly	MEDIUM	Slim dart body, double wing surfaces — agile and odd.	48

Classic Dart

The one every pilot learns first — fast, straight and true.

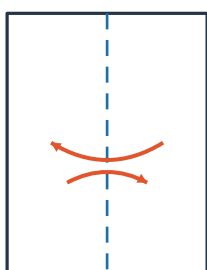
The original schoolyard jet. Five folds, two wings, zero excuses. Its narrow swept wings trade lift for speed, so it flies fast, straight lines and points like a thrown spear. Master this one first: nearly every other design builds on its moves.

SPEED	■ ■ ■ ■ ■
DISTANCE	■ ■ ■ ■ ■
AIRTIME	■ ■ ■ ■ ■
STUNTS	■ ■ ■ ■ ■

FOLD TIME ~2 MIN

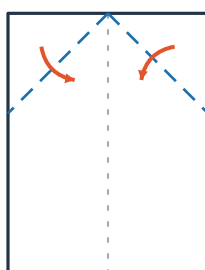
PAPER: Standard letter or A4 printer paper.

1



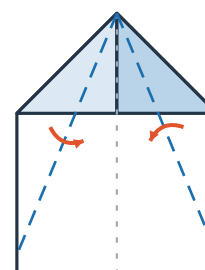
fold in half lengthwise, then unfold

2



fold the top corners to the center

3



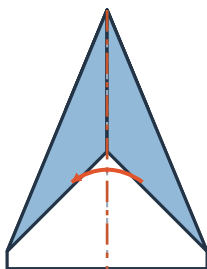
fold the angled edges to the center

Fold the sheet in half lengthwise (long edge to long edge), crease firmly, then unfold. This center crease is your guide line for everything that follows.

Fold the top two corners in so their top edges lie exactly along the center crease. The corners should meet in the middle, forming a point at the top.

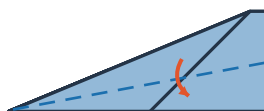
Fold each angled edge in to the center line again, keeping the nose point sharp. The plane is now a long, slim kite shape.

4



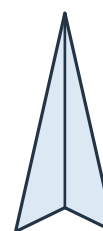
fold the plane in half away from you

5



fold the wing down; flip and repeat

6



ready for flight

Fold the whole plane in half away from you (mountain fold) along the original center crease, so the flaps you just made are hidden inside.

Make the first wing: fold the top layer down so the slanted top edge lines up with the bottom of the fuselage. Flip the plane over and fold the second wing to match exactly.

Open the wings so they sit level, forming a slight 'Y' with the body. Pinch the fuselage under the center of gravity and throw firmly, straight ahead.

FIELD NOTES

- Symmetry beats strength: identical wings matter more than a hard throw.
- Sharpen every crease with a fingernail or the edge of a ruler.
- Diving? Bend the back edges of both wings up 2–3 mm.

TRIM: Throw flat and hard at shoulder height. For range, aim slightly upward at about 10–15 degrees.

The Arrow

A dart with winglets — straighter than straight.

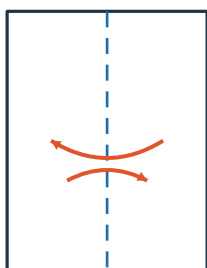
Take the Classic Dart, add vertical wingtips, and you get the Arrow: a precision instrument that resists rolling and tracks dead straight down a hallway. The winglets act like the tail fins on a real arrow, damping out wobble mid-flight.

SPEED	■ ■ ■ ■ ■
DISTANCE	■ ■ ■ ■ ■
AIRTIME	■ ■ ■ ■ ■
STUNTS	■ ■ ■ ■ ■

FOLD TIME ~3 MIN

PAPER: Standard letter or A4 printer paper.

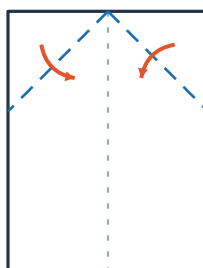
1



fold in half lengthwise, then unfold

Fold the sheet in half lengthwise and unfold to set the center guide crease.

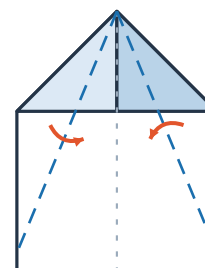
2



fold the top corners to the center

Fold both top corners to the center crease so the edges meet in the middle.

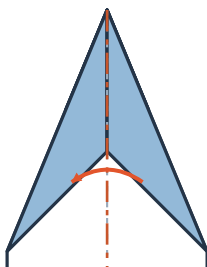
3



fold the angled edges to the center line again

Fold the angled edges to the center line a second time for a slim, pointed profile.

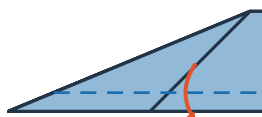
4



fold the plane in half away from you

Mountain-fold the plane in half away from you along the center crease.

5



wing creases parallel to the bottom

Fold each wing down, leaving about 2 cm (3/4 in) of fuselage below. Keep both wing creases parallel to the bottom edge.

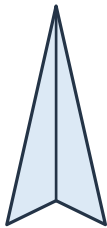
6



fold the wingtips straight up

Fold the outer 1.5 cm (1/2 in) of each wingtip straight up at a right angle. These vertical winglets keep the Arrow from rolling.

7



ready for flight

Level the wings, square up the winglets, and launch hard and flat. The Arrow rewards a confident throw.

FIELD NOTES

- Winglets must be the same height or the plane carves a curve.
- Great indoor accuracy plane — set up a target and keep score.

TRIM: If it veers left, slightly enlarge the right winglet (or angle it outward) and vice versa.

Bulldog Dart

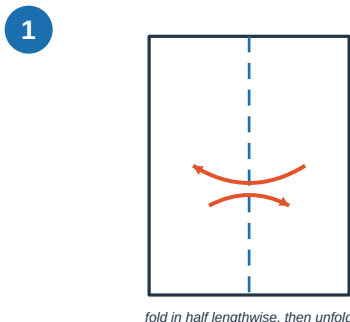
A blunt-nosed brawler that survives every crash landing.

The Bulldog trades the needle nose for a folded-back snub that loads weight up front and shrugs off wall strikes. It is the perfect second plane: one new move (the nose tuck) and a much more durable, better-balanced flyer.

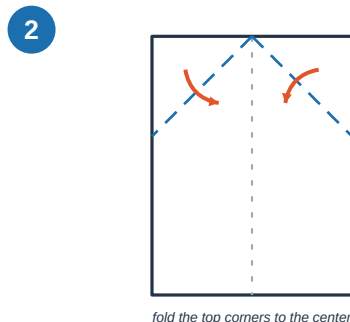
SPEED	■ ■ ■ ■ ■
DISTANCE	■ ■ ■ ■ ■
AIRTIME	■ ■ ■ ■ ■
STUNTS	■ ■ ■ ■ ■

FOLD TIME ~3 MIN

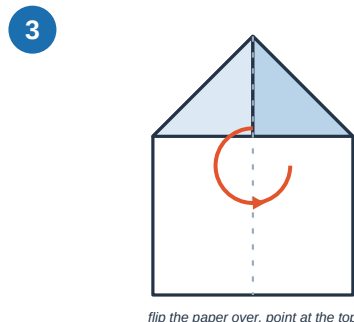
PAPER: Standard letter or A4 printer paper.



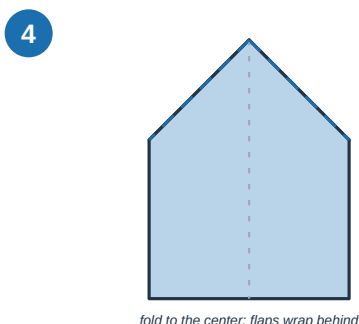
Fold in half lengthwise and unfold for the center crease.



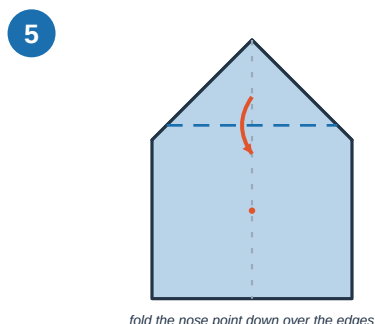
Fold the top corners in to the center crease.



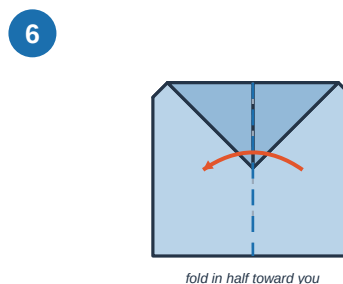
Flip the paper over, keeping the point at the top.



Fold the top angled sides to the center line again. The flaps from step 2 will wrap around from behind — that's what packs weight into the nose.

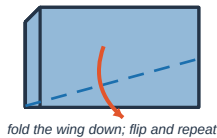


Fold the nose point down (toward you) over the folded edges, so the tip sticks out about 1 cm past the straight edge. This blunt snub is the bulldog's face.

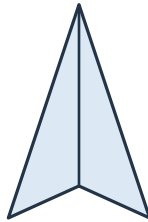


Fold the plane in half toward you along the center crease, keeping the snub nose on the outside.

7



8



Fold each wing down so its top edge lines up with the flat bottom of the fuselage. Flip and repeat on the other side.

Level the wings and give it a smooth, medium-strength throw. Too hard and the Bulldog climbs, stalls and dive-bombs.

FIELD NOTES

- This is the best plane to learn throwing technique: it's honest about a bad release.
 - The snub nose protects the folds — a Bulldog lasts dozens of flights.
- TRIM:** Stalling (climb-then-drop)? Throw softer or curl the wing trailing edges down a touch.

The Concorde

Supersonic looks, hallway-dogfight handling.

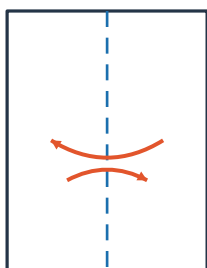
Long, low and pointed like its namesake, the Concorde is a three-stage dart: each fold to the center makes the nose denser and the profile sleeker. It won't float, but for raw speed in a straight line and razor turns off a flicked wrist, nothing in this hangar matches it.

SPEED	■ ■ ■ ■ ■
DISTANCE	■ ■ ■ ■ ■
AIRTIME	■ ■ ■ ■ ■
STUNTS	■ ■ ■ ■ ■

FOLD TIME ~4 MIN

PAPER: Standard or slightly heavier paper; crisp creases are everything.

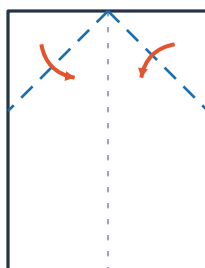
1



fold in half lengthwise, then unfold

Fold lengthwise, crease hard, unfold.

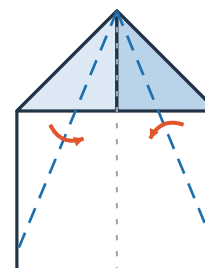
2



fold the top corners to the center

Fold the top corners to the center crease.

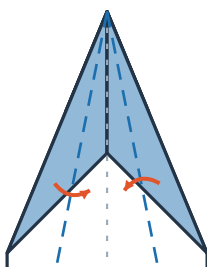
3



fold the angled edges to the center again

Fold the angled edges to the center line a second time.

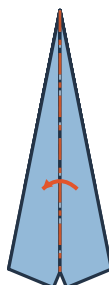
4



fold the edges to the center a third time

Fold the angled edges to the center a third time. The layers are thick now: work slowly and keep the nose tip needle-sharp.

5



fold the plane in half away from you

Mountain-fold the plane in half away from you.

6



slim wings: ~1.5 cm of fuselage

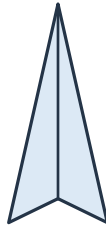
Fold narrow wings: crease each side down leaving only about 1.5 cm of fuselage. The Concorde flies on slim delta wings.

7



fold up small 1 cm winglets

8



ready for flight

Fold up small 1 cm winglets at each tip for tracking.

Wings level, fins vertical. Throw hard and flat — this plane wants energy.

FIELD NOTES

- Heavier 28 lb paper makes the dense nose even better.
- Indoors, throw slightly downward from high — it accelerates.

TRIM: A tiny up-bend on one wing's trailing edge makes it carve repeatable combat turns.

The Sprinter

A strip-loaded nose for drag-race acceleration.

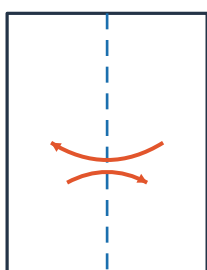
Instead of folding corners for nose weight, the Sprinter rolls the top edge down in tight strips — the same ballast trick used by competition distance planes — then finishes like a dart. The result punches through air with almost no flutter.

SPEED	■ ■ ■ ■ ■
DISTANCE	■ ■ ■ ■ ■
AIRTIME	■ ■ ■ ■ ■
STUNTS	■ ■ ■ ■ ■

FOLD TIME ~4 MIN

PAPER: Lightweight printer paper folds the strips most cleanly.

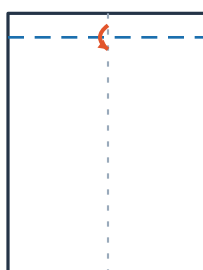
1



fold in half lengthwise, then unfold

Fold lengthwise and unfold for the center guide.

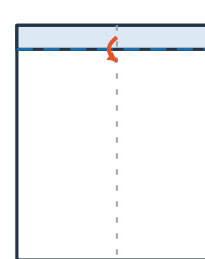
2



fold the top edge down 2.5 cm

Fold the top edge down about 2.5 cm (1 in) and crease hard.

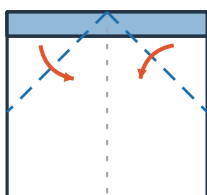
3



fold the band down twice more

Fold that band down again, and once more — three tight strips total. You've built a stiff, heavy leading bar across the top.

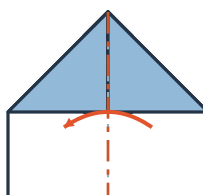
4



fold the corners down to the center

Fold the top corners (through all the strip layers) down to the center crease, forming a point. Press hard — you're creasing several layers.

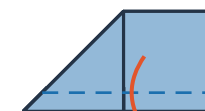
5



fold the plane in half away from you

Fold the plane in half away from you along the center crease.

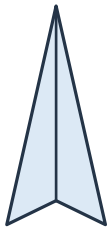
6



wing creases parallel to the bottom

Fold both wings down, leaving about 2 cm of fuselage, keeping the creases parallel to the bottom.

7



ready for flight

Level the wings and release flat and fast. The dense nose bar keeps the Sprinter on rails.

FIELD NOTES

- Crease the strips over a table edge for crisp, flat rolls.
- This nose survives nose-first landings better than any dart.

TRIM: Nose-heavy dive? Add a whisper of up-elevator at the wing's back edge.

Nakamura Lock

The locked-nose legend — the best easy plane ever folded.

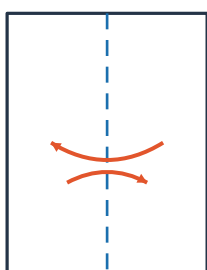
Known in classrooms worldwide as 'the good one.' A small tip fold locks the nose flaps shut so the plane never blossoms open mid-flight. Effortless to fold, shockingly far-flying, and the single best return on two minutes of folding you will find.

SPEED	■ ■ ■ ■ ■
DISTANCE	■ ■ ■ ■ ■
AIRTIME	■ ■ ■ ■ ■
STUNTS	■ ■ ■ ■ ■

FOLD TIME ~4 MIN

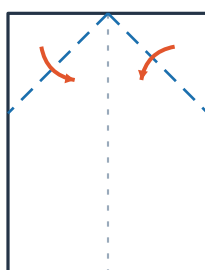
PAPER: Standard letter or A4 printer paper.

1



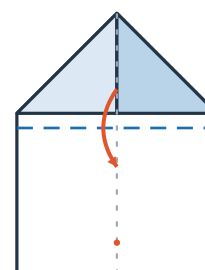
fold in half lengthwise, then unfold

2



fold the top corners to the center

3



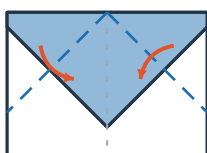
fold the point down; tip near the bottom

Fold the sheet in half lengthwise and unfold.

Fold the top corners in to the center crease.

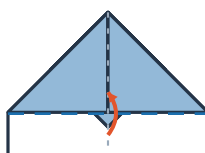
Fold the whole pointed top straight down, so the tip lands on the center crease about 2.5 cm (1 in) above the bottom edge.

4



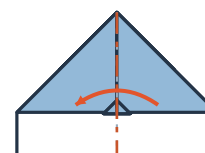
corners to the center, tip peeking below

5



fold the small tip UP: the lock

6



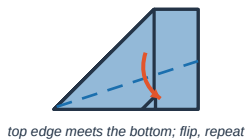
fold in half; lock on the outside

Fold the new top corners in to the center line so they meet just above the buried tip — a small triangle of the tip should still peek out below them.

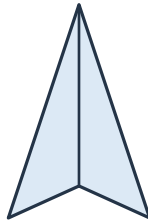
Fold that little peeking tip UP over both flaps. This is the famous lock: it pins everything shut.

Fold the plane in half away from you so the lock stays on the outside of the fuselage.

7



8



Fold each wing down so the straight top edge meets the bottom of the fuselage; flip and match the second wing.

Wings level, lock underneath. A smooth medium throw gives long, flat, satisfying flights every single time.

FIELD NOTES

- If you teach one plane to a kid, teach this one.
 - The lock only works if the corners in step 4 don't overlap the tip — leave it room to fold up.
- TRIM:** It self-corrects most errors. For max range, throw at 30–40° with medium-hard power.

The Suzanne

Folded in the style of the world-record distance glider.

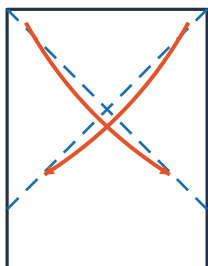
In 2012, a design like this — John Collins' 'Suzanne,' thrown by Joe Ayoob — flew 226 feet 10 inches to take the Guinness distance record, the first glider ever to do it. Unlike darts, it launches hard, then transitions into a true glide. This tribute version captures its key ideas: diagonal-locked nose weight, wide wings, and a variable dihedral.

SPEED	■ ■ ■ ■ ■
DISTANCE	■ ■ ■ ■ ■
AIRTIME	■ ■ ■ ■ ■
STUNTS	■ ■ ■ ■ ■

FOLD TIME ~8 MIN

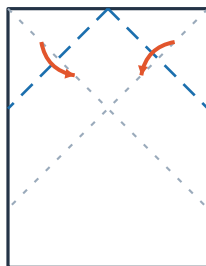
PAPER: Use crisp, light 20 lb (75–80 gsm) paper. Precision matters more than on any other design here.

1



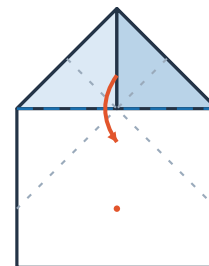
crease both diagonals, then unfold

2



corners in: points land on the creases

3



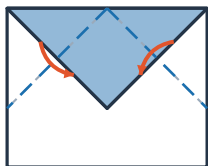
fold the top down through the X center

Fold the top-right corner down so the top edge lies along the left side; crease and unfold. Repeat with the top-left corner to the right side. You now have an X of creases in the top square.

Fold each top corner in so its point lands exactly on the opposite diagonal crease, edges meeting near the center line. Take your time — equal sides are everything.

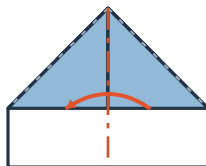
Fold the whole top section down, creasing right through the center of the X. The folded corners are now trapped inside — this is the weighted nose bar.

4



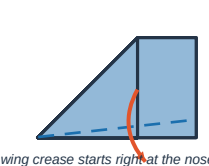
corners to the center through all layers

5



fold the plane in half away from you

6



wing crease starts right at the nose tip

Fold the top corners to the center line through all layers, forming the nose point. Crease emphatically — a bone folder or ruler edge helps.

Fold the plane in half away from you along the center line.

Fold each wing down with a crease that starts right at the nose tip and leaves about 2 cm of fuselage at the tail. Match the second wing exactly.

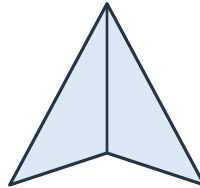
7



front view: a soft upward Y

Front view check: the wings should angle slightly upward in a soft 'Y'. The record plane uses this dihedral to stay stable at launch speed and lift in the glide.

8



launch hard at 40 degrees

Grip near the nose, launch hard at about 40 degrees. A proper Suzanne-style flight climbs, levels off, and glides in for a soft landing. (The record team used small bits of tape on nose, spine and wing — allowed under Guinness rules and worth trying.)

FIELD NOTES

- Three tiny tabs of tape — nose, mid-spine, tail — transform its consistency.
- Crease accuracy in step 2 decides whether it flies straight; refold rather than accept a skewed nose.

TRIM: Test-glide gently first. Dive = add slight up-bend at the wing trailing edges; stall = flatten them out.

The Hammer

A dense, armored nose for maximum outdoor range.

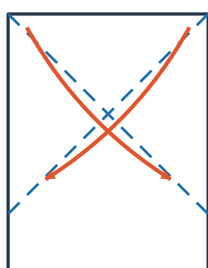
The Hammer concentrates an extraordinary amount of paper into its first few centimeters, then hangs broad wings off the back. It punches through breeze that scatters lighter planes, making it the best outdoor distance flyer in this collection.

SPEED	■ ■ ■ ■ ■
DISTANCE	■ ■ ■ ■ ■
AIRTIME	■ ■ ■ ■ ■
STUNTS	■ ■ ■ ■ ■

FOLD TIME ~8 MIN

PAPER: Standard paper; heavier 24 lb works outdoors.

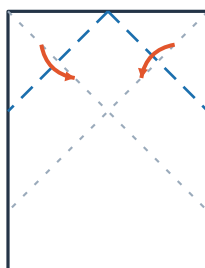
1



crease both diagonals, then unfold

Fold and unfold both diagonals of the top square to make an X of creases.

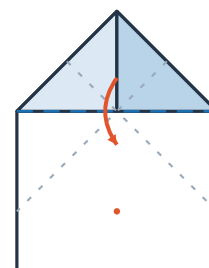
2



fold the top corners to the center

Fold the top corners in to the center crease.

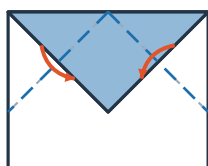
3



fold the point down onto the center line

Fold the pointed top straight down over the flaps, tip landing on the center crease.

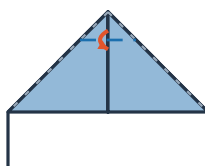
4



corners in over the buried point

Fold the new top corners in to the center line over the buried point.

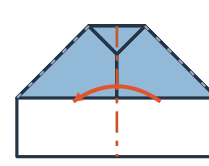
5



fold the whole top edge down ~3 cm

Fold the entire reinforced top edge down one more time, about 3 cm. The nose is now a solid multi-layer slab — the hammer head.

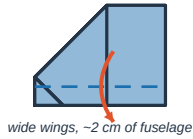
6



fold the plane in half away from you

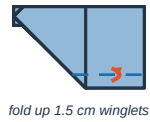
Fold the plane in half away from you along the center line, keeping the nose slab outside.

7



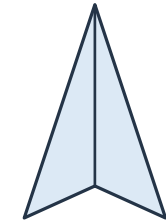
Fold wide wings down on each side, leaving about 2 cm of fuselage.

8



Fold up 1.5 cm winglets at each tip. Outdoors these keep the Hammer from being rolled by gusts.

9



launch hard at ~40 degrees

Take it outside, face the open field, and launch hard at about 40 degrees. The Hammer converts arm strength into distance better than anything else here.

FIELD NOTES

- Crease the nose slab over a table edge — your fingers alone can't fully compress it.
- Light breeze in your face actually helps; launch into it.

TRIM: If it corkscrews, the wing creases aren't symmetric — refold the lighter wing rather than bending fixes into it.

The Eagle

A Toda-style competition dart, tuned for clean release.

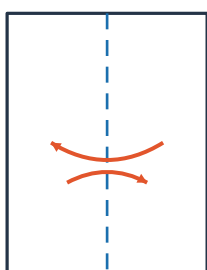
Japanese champion Takuo Toda's record planes are deceptively simple darts with meticulous balance — his designs have flown over 60 meters and held the world duration record. The Eagle follows that school: strip-weighted nose, modest wings, and everything tuned around a clean, hard release.

SPEED	■ ■ ■ ■ ■
DISTANCE	■ ■ ■ ■ ■
AIRTIME	■ ■ ■ ■ ■
STUNTS	■ ■ ■ ■ ■

FOLD TIME ~5 MIN

PAPER: Light, stiff paper; A4 is closest to competition stock.

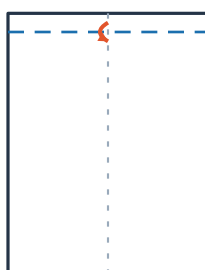
1



fold in half lengthwise, then unfold

Fold lengthwise, crease hard, unfold.

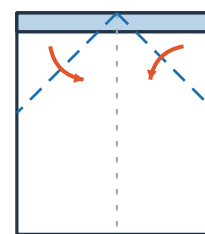
2



fold the top edge down 2 cm, twice

Fold the top edge down 2 cm, then fold that band down again — two crisp strips.

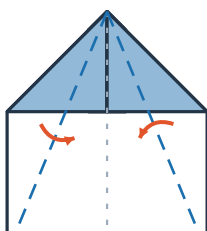
3



corners to the center through the strips

Fold the top corners down to the center crease, forming the nose point through the strip layers.

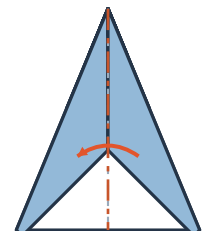
4



fold the angled edges to the center again

Fold the angled edges in to the center line once more for a narrow profile.

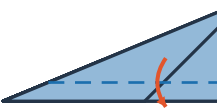
5



fold the plane in half away from you

Fold in half away from you along the center crease.

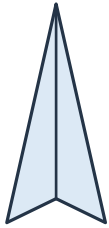
6



slim wings; edges align when folded

Fold slim wings, leaving about 2 cm of fuselage; the wing's top edges should align perfectly when folded.

7



full power, 45 degrees up

Hold at the balance point (about a third back from the nose), and launch at 45 degrees with full power and a flat wrist. Toda's secret is the release: the plane must leave your hand with zero sideways twist.

FIELD NOTES

- Practice the throw separately: a perfect Eagle with a twisting release flies worse than a sloppy dart thrown clean.
- Indoors in still air this design holds long, flat lines.

TRIM: Add the faintest up-curl on both trailing edges for the climb-and-glide competition profile.

Alpha Strike

A locked nose plus tall winglets — far and forgiving.

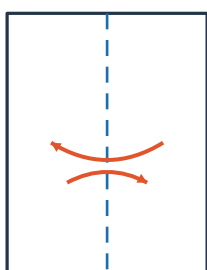
A souped-up cousin of the Nakamura Lock. The nose lock keeps it from opening, while taller winglets and a slightly shorter nose fold shift the balance for longer, flatter cruises. One of the most consistent 'far on every throw' planes you can fold.

SPEED ■■■■ ■
 DISTANCE ■■■■ ■
 AIRTIME ■■■■ ■
 STUNTS ■ ■■ ■■ ■■

FOLD TIME ~5 MIN

PAPER: Standard letter or A4 printer paper.

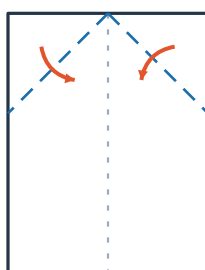
1



fold in half lengthwise, then unfold

Fold in half lengthwise and unfold.

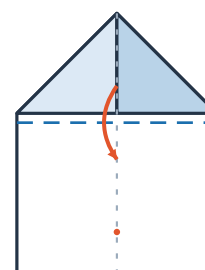
2



fold the top corners to the center crease

Fold the top corners to the center crease.

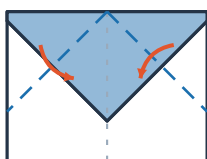
3



fold the point down two-thirds of the way

Fold the point down about two-thirds of the way to the bottom — shorter than the Nakamura. The nose will sit further back.

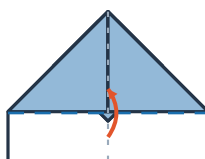
4



corners in; tip stays exposed below

Fold the top corners to the center, leaving the buried tip exposed below them.

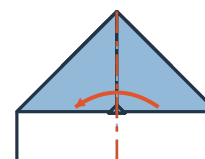
5



fold the exposed tip up: the lock

Fold the exposed tip up over the flaps to lock the nose.

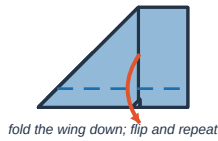
6



fold in half; lock on the outside

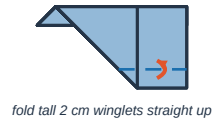
Fold in half away from you, lock on the outside.

7



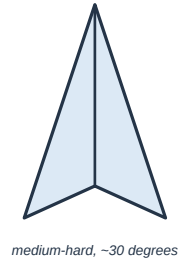
Fold the wings down leaving ~2 cm of fuselage.

8



Fold tall 2 cm winglets straight up at each tip. They should stand perfectly vertical and mirror each other.

9



Launch medium-hard at about 30 degrees. The Alpha settles into a long flat cruise and lands gently.

FIELD NOTES

- The most forgiving distance plane here — great for competitions with mixed skill levels.
- TRIM:** Drifting in circles? One winglet is leaning — square both to true vertical.

Stealth Glider

The hang-time champion — throw it high and count the seconds.

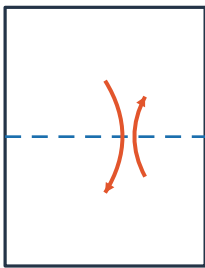
Wide delta wings and a tucked, locked nose let the Stealth Glider fly astonishingly slowly. Hurl it straight up, let it flip over the top, and it settles into a creeping flat glide. With patient tuning this is the longest-flying plane in the hangar.

SPEED	
DISTANCE	
AIRTIME	
STUNTS	

FOLD TIME ~6 MIN

PAPER: Light paper floats best; phonebook-thin paper is magic.

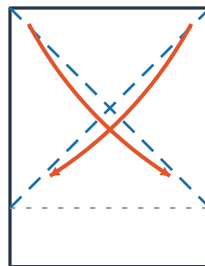
1



fold in half the short way, then unfold

Fold the sheet in half the short way (top edge to bottom edge) and unfold.

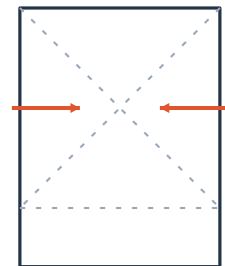
2



crease both diagonals of the top square

Fold and unfold both diagonals in the top square to form an X.

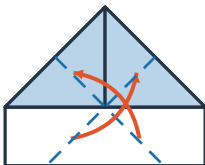
3



push sides in; top collapses into a triangle

Push the left and right sides of the X inward along the creases and flatten: the top of the sheet collapses into a layered triangle (a 'waterbomb base').

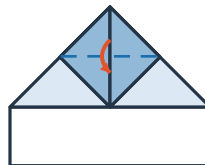
4



fold the front corners up to the peak

Take the front-layer bottom corners of the triangle and fold them up to the peak, forming a small diamond on top.

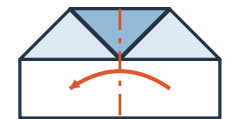
5



fold the peak down: locks the nose

Fold the peak of that diamond down over the two flaps, tucking it snug. This locks the nose shut and loads it with weight.

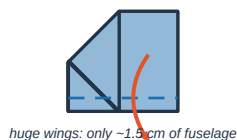
6



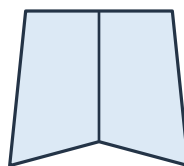
gently fold in half away from you

Gently fold the plane in half away from you. Don't crush the nose lock — a soft spine crease is fine.

7



8



Fold enormous wings: crease each side down leaving only about 1.5 cm of fuselage. Nearly all the paper becomes wing.

Flatten the wings level. Now the unusual part: throw it STRAIGHT UP as hard as you can. It tumbles over the top, recovers, and begins its long slow ride down.

FIELD NOTES

- Time-aloft contests are won on launch height — it's a glider strapped to your arm strength.
- Tiny trailing-edge bends change everything; adjust in millimeters.

TRIM: Spiraling? Level the wings and remove any twist. Diving? A breath of up-elevator on both wings.

Sea Glider

A moth-winged delta that drifts straight and far.

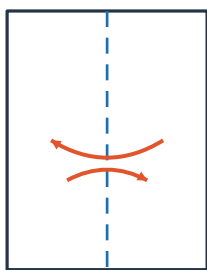
A tucked nose and one huge delta wing surface give the Sea Glider the silhouette of a giant moth and the manners of a seabird: launched level, it locks onto a heading and drifts far past where darts drop.

SPEED	
DISTANCE	
AIRTIME	
STUNTS	

FOLD TIME ~5 MIN

PAPER: Standard paper; slightly heavy paper improves its line outdoors.

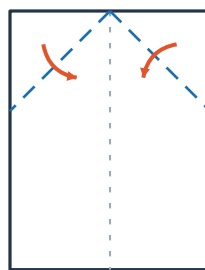
1



fold in half lengthwise, then unfold

Fold in half lengthwise and unfold.

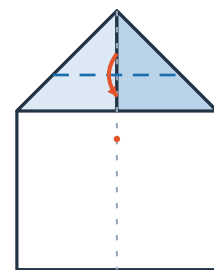
2



fold the top corners to the center

Fold the top corners to the center crease.

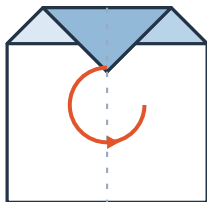
3



fold the point down to mid-sheet

Fold the pointed top down so the tip lands on the center line about halfway down the sheet.

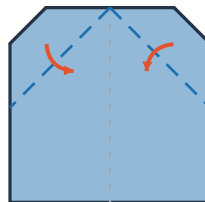
4



flip the paper over, point at the top

Flip the paper over, folded point underneath.

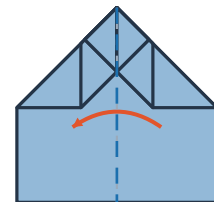
5



corners in: the nose wraps around, locking

Fold the top corners to the center line. As you do, the nose point wraps around from behind, locking the layers together.

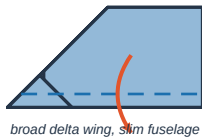
6



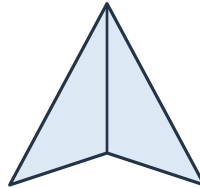
fold in half toward you

Fold the plane in half toward you along the center crease.

7



8



Fold each wing down leaving a slim ~1.5 cm fuselage; the broad delta wing is the whole point.

Wings level with a hint of upward angle, then a smooth, level, medium launch. The Sea Glider does the rest.

FIELD NOTES

- Launched from a balcony or hilltop it seems to fly forever.
- Looks fantastic folded from patterned paper — the wing is one big canvas.

TRIM: Wandering? Add 1 cm folded-up wingtips for heading stability.

The Albatross

The all-purpose floater — calm, stable, dependable.

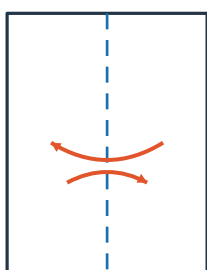
A heavy nose, generous wings and tall winglets make the Albatross the station wagon of paper aviation: it does everything acceptably and nothing badly. Ideal for younger pilots graduating from the dart, and happiest indoors.

SPEED	
DISTANCE	
AIRTIME	
STUNTS	

FOLD TIME ~4 MIN

PAPER: Medium to heavyweight paper (24–28 lb) is recommended.

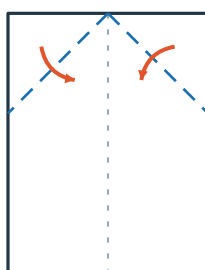
1



fold in half lengthwise, then unfold

Fold in half lengthwise and unfold.

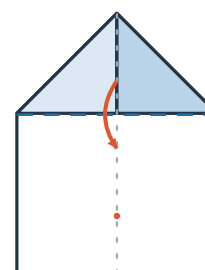
2



fold the top corners to the center crease

Fold the top corners to the center crease.

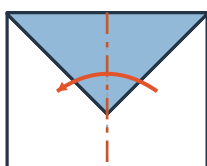
3



deep nose fold: tip ~6 cm from the bottom

Fold the pointed top down over the flaps so the tip lands about 6 cm above the bottom edge — a deep, nose-heavy tuck.

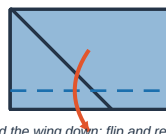
4



fold the plane in half away from you

Fold the plane in half away from you.

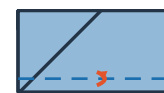
5



fold the wing down; flip and repeat

Fold both wings down, leaving about 2 cm of fuselage.

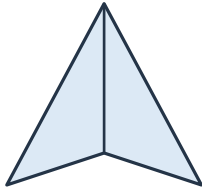
6



fold up 1.5 cm vertical stabilizers

Fold up 1.5 cm vertical stabilizers at each wingtip.

7



easy, level toss

Level everything and launch with an easy, level toss. The Albatross cruises straight ahead and sets down softly.

FIELD NOTES

- Bend the back of the wings up slightly and throw harder for lazy acrobatic loops — it's secretly playful.

TRIM: With heavier paper, all adjustments need to be slightly larger to take effect.

The Moth

Three minutes to fold, all wing, pure float.

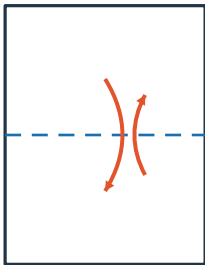
The simplest glider in the collection: a strip-weighted leading edge and two giant square wings. The Moth flies slowly enough to walk beside, making it perfect for small rooms, toddlers' first catches, and quiet desk-side test flights.

SPEED	
DISTANCE	
AIRTIME	
STUNTS	

FOLD TIME ~3 MIN

PAPER: The lighter the better.

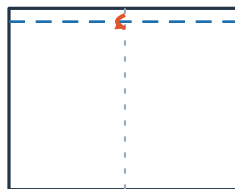
1



crease the short way: this is the spine

Fold the paper in half the short way and unfold — this center crease will be the spine.

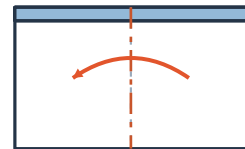
2



roll the top edge down in three strips

Turn the paper sideways (long edge at top). Fold the top edge down about 1.5 cm; repeat twice more, rolling a stiff bar along the leading edge.

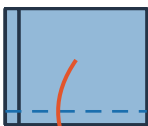
3



fold in half; bar on the outside

Fold in half away from you along the spine crease, keeping the rolled bar on the outside.

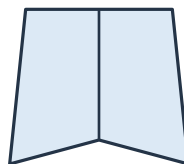
4



almost everything becomes wing

Fold each wing down leaving only ~1.5 cm of fuselage — almost everything is wing.

5



the gentlest push, nose a touch down

Wings flat and level. Release it with the gentlest push, nose very slightly down, and let it carry itself.

FIELD NOTES

- Try dropping it from a stairwell — no throw at all.
- A great canvas for decorating: the wing is one open square.

TRIM: Floaters hate asymmetry: if it turns, your fuselage halves aren't even.

Sky King

A duration specialist in the spirit of the 29-second record.

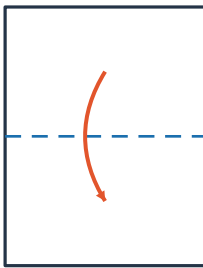
Takuo Toda's 'Sky King' held the world record for time aloft — 29.2 seconds of flight from a single throw. Duration planes are their own art: short, wide, nose-dense, and trimmed within a millimeter of stalling. This tribute follows that recipe.

SPEED ■■■■■
 DISTANCE ■■■■■
 AIRTIME ■■■■■
 STUNTS ■■■■■

FOLD TIME ~8 MIN

PAPER: Stiff, light paper. Record planes use special stock; light A4 is the closest household match.

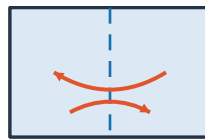
1



fold in half and LEAVE it folded

Fold the sheet in half the short way and leave it folded — the Sky King flies on a doubled, half-size sheet for stiffness.

2



crease a center line, then unfold

With the folded edge at the top, fold in half lengthwise and unfold for a center crease.

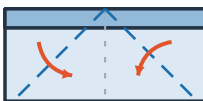
3



fold the doubled edge down 2 cm, twice

Fold the doubled top edge down 2 cm, then again — a very dense four-plus-layer nose bar.

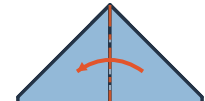
4



corners to the center through all layers

Fold the top corners to the center line through all layers. Press creases with a ruler edge — finger pressure isn't enough.

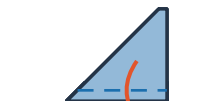
5



fold the plane in half away from you

Fold in half away from you along the center crease.

6



thin fuselage: duration lives in wing area

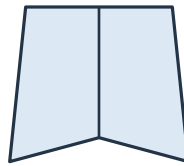
Fold the wings with a thin ~1.2 cm fuselage; duration lives in wing area.

7



slight dihedral; curl trailing edges up

8



launch straight up, full power

Set a slight upward dihedral and curl the last centimeter of each trailing edge up just a hair — the stall-edge trim that duration flying depends on.

Launch straight up at full power. A trimmed Sky King noses over at the top and descends in slow, flat circles. Count the seconds; 10+ indoors means you've trimmed it well.

FIELD NOTES

- Real duration attempts happen indoors — gyms and atriums — where no breeze can interfere.
- Adjust trim between every flight; champions never stop tuning.

TRIM: It should fly on the edge of a stall: too much up-trim and it porpoises, too little and it dives. Split the difference in tiny steps.

The Condor

A big-winged soarer that loves a high launch point.

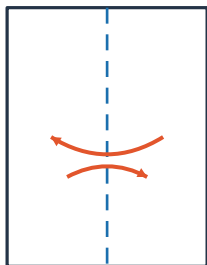
The Condor wraps its nose weight in a double corner fold, then spends the rest of the sheet on wingspan. From a balcony, stairwell or hilltop it converts every foot of height into yards of forward drift.

SPEED	■ ■ ■ ■ ■
DISTANCE	■ ■ ■ ■ ■
AIRTIME	■ ■ ■ ■ ■
STUNTS	■ ■ ■ ■ ■

FOLD TIME ~5 MIN

PAPER: Standard printer paper.

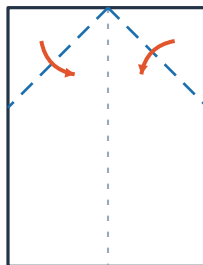
1



fold in half lengthwise, then unfold

Fold in half lengthwise and unfold.

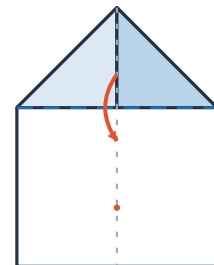
2



fold the top corners to the center crease

Fold the top corners to the center crease.

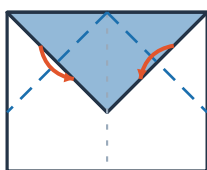
3



fold down along the flap bottoms

Fold the pointed section down so the fold line runs along the bottom of the corner flaps.

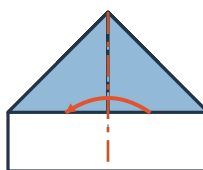
4



corners in over the buried point

Fold the new top corners in to the center line over the buried point — the doubled nose is the Condor's engine.

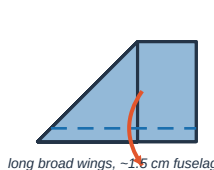
5



fold the plane in half away from you

Fold in half away from you.

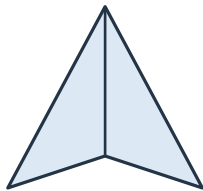
6



long broad wings, ~1.5 cm fuselage

Fold long broad wings leaving ~1.5 cm of fuselage.

7



launch level from height

Find height — stairs, deck, playground tower
— and launch level with medium power.
Watch it stretch its glide.

FIELD NOTES

- In light outdoor thermals off warm pavement, a Condor can visibly extend its glide — real soaring physics in paper.

TRIM: Trim for the flattest possible glide before going to the high launch spot.

The Stunt Plane

Loops on demand — bend the elevators and let it rip.

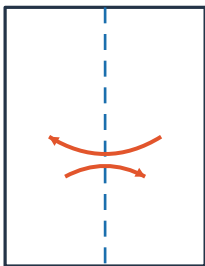
A dense-nosed mid-wing design with big control surfaces. Bend the wing trailing edges up hard and a strong throw sends it around a full vertical loop; flatten them and it flies straight. One plane, a dozen tricks, all in how you bend the paper.

SPEED	■ ■ ■ ■ ■
DISTANCE	■ ■ ■ ■ ■
AIRTIME	■ ■ ■ ■ ■
STUNTS	■ ■ ■ ■ ■

FOLD TIME ~5 MIN

PAPER: Standard paper; it needs some flex in the control surfaces.

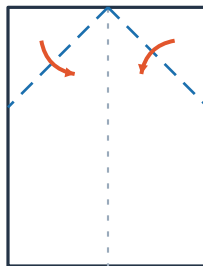
1



fold in half lengthwise, then unfold

Fold in half lengthwise and unfold.

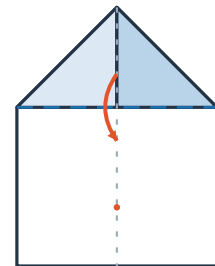
2



fold the top corners to the center crease

Fold the top corners to the center crease.

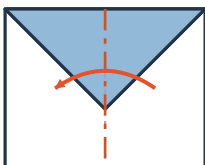
3



fold the point down to load the nose

Fold the point down over the flaps to load the nose.

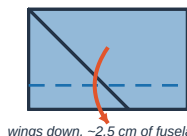
4



fold the plane in half away from you

Fold in half away from you.

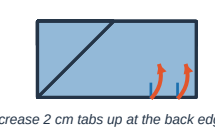
5



wings down, ~2.5 cm of fuselage

Fold both wings down, leaving about 2.5 cm of fuselage.

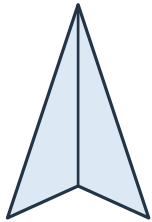
6



crease 2 cm tabs up at the back edges

At the back edge of each wing, make two short 1 cm cuts-free bends: just crease and bend a 2 cm-wide tab upward about 45°. These are your elevators.

7



elevators up + hard throw = loop

Elevators up + a hard, slightly upward throw = a full loop. Throw straight down from height with elevators up and it swoops out and returns level. Endless experiments await.

FIELD NOTES

- One elevator up, one flat = barrel rolls.
 - Bend, don't tear: tabs survive dozens of re-adjustments.
- TRIM:** Bigger bends, tighter loops — until it stalls. Find the edge.

The Boomerang

Throw it away and it comes home.

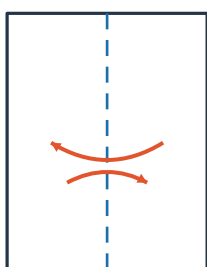
The party trick of paper aviation. Asymmetric trim makes the Boomerang carve a wide banked circle: thrown at the right angle it arcs across the room and returns to your hand — or at least your general vicinity. Expect to tune; the payoff is worth it.

SPEED	
DISTANCE	
AIRTIME	
STUNTS	

FOLD TIME ~7 MIN

PAPER: Standard letter or A4 printer paper.

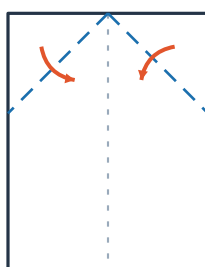
1



fold in half lengthwise, then unfold

Fold in half lengthwise and unfold.

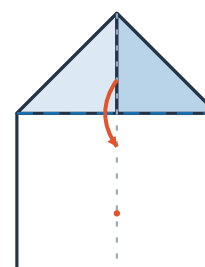
2



fold the top corners to the center crease

Fold the top corners to the center crease.

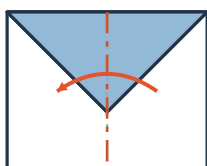
3



fold the point down for nose weight

Fold the point down over the flaps for nose weight.

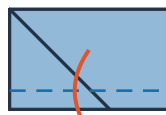
4



fold the plane in half away from you

Fold in half away from you.

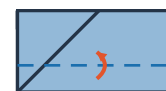
5



fold the wing down; flip and repeat

Fold both wings down leaving ~2 cm of fuselage.

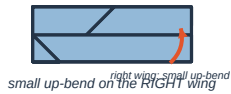
6



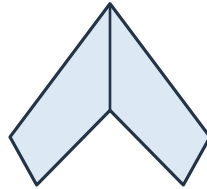
LEFT wingtip up at about 45 degrees

Fold the outer 3 cm of the LEFT wingtip up at about 45° (if you're right-handed). This asymmetry is the boomerang's steering.

7



8



Add a small upward bend at the back edge of the RIGHT wing. Together the two tweaks make it climb and bank left continuously.

Hold the plane tilted about 45° to the right and throw firmly, upward at about 45°. It should carve up and left in a big circle back toward you. Adjust the tip and elevator until the circle closes. Lefties: mirror everything.

FIELD NOTES

- High ceilings are everything — gyms, garages, living rooms with vaulted ceilings.
- Catching your own throw is a legitimate party trick. Practice in private, debut in public.

TRIM: Circle too wide? More elevator. Crashes mid-turn? Less wingtip angle. Change one variable per throw.

The Spinner

A corkscrewing dart that drills through the air.

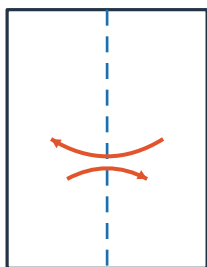
Deliberately twisted wingtips make the Spinner roll continuously as it flies — a smooth aileron roll the whole way down its line. Useless for distance, hypnotic to watch, and a lesson in roll physics you can fold in four minutes.

SPEED ■■■■ ■
 DISTANCE ■■■ ■■
 AIRTIME ■■■ ■■
 STUNTS ■■■■ ■■

FOLD TIME ~4 MIN

PAPER: Standard paper.

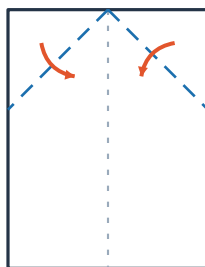
1



fold in half lengthwise, then unfold

Fold in half lengthwise and unfold.

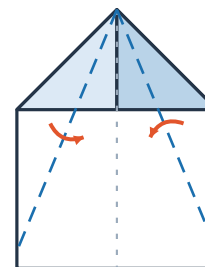
2



fold the top corners to the center crease

Fold the top corners to the center crease.

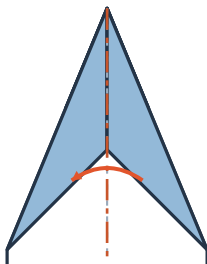
3



fold the angled edges to the center again

Fold the angled edges to the center once more — standard dart body.

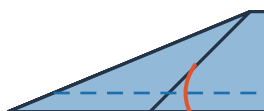
4



fold the plane in half away from you

Fold in half away from you.

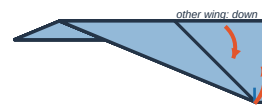
5



fold the wing down; flip and repeat

Fold both wings down leaving ~2 cm of fuselage.

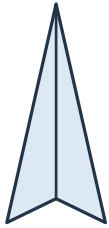
6



one back corner up 30°, the other down 30°

Here's the trick: bend the back corner of ONE wingtip up about 30°, and the same spot on the OTHER wing down 30°. Opposite twists = continuous roll.

7



hard and flat: instant corkscrew

Throw hard and flat. The Spinner immediately starts its corkscrew and holds it across the room.

FIELD NOTES

- More twist = faster spin, shorter flight. Find your favorite ratio.
- Color one wing with a marker to make the roll easier to see.

TRIM: If it spirals into the ground instead of rolling on a line, reduce both twists equally.

The Loop

The easiest trick plane — a guaranteed vertical loop.

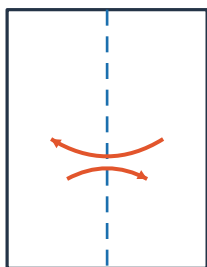
Short, blunt and heavily trimmed, the Loop exists to do exactly one thing: fly a complete vertical circle and land at your feet. It's the fastest 'wow' in this collection and a perfect first stunt plane for kids.

SPEED ■■■■■
 DISTANCE ■■■■■
 AIRTIME ■■■■■
 STUNTS ■■■■■

FOLD TIME ~3 MIN

PAPER: Standard paper.

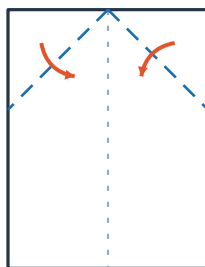
1



fold in half lengthwise, then unfold

Fold in half lengthwise and unfold.

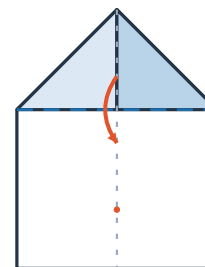
2



fold the top corners to the center crease

Fold the top corners to the center crease.

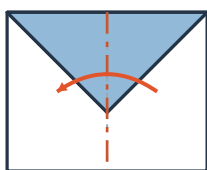
3



short, stubby, nose-heavy body

Fold the point down over the flaps — a short, stubby, nose-heavy body.

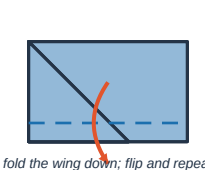
4



fold the plane in half away from you

Fold in half away from you.

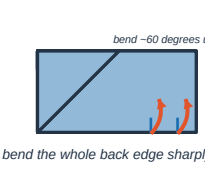
5



fold the wing down; flip and repeat

Fold the wings down leaving about 2 cm of fuselage.

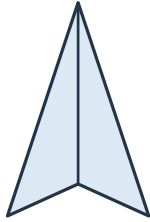
6



bend the whole back edge sharply up

Bend the entire back 2 cm of BOTH wings sharply upward — about 60°. Yes, that much. This plane is all elevator.

7



point straight up and throw hard

Point it straight up and throw hard. It climbs, flips over backward, dives, pulls out — a complete loop. Throwing level instead gives a climb-and-return boomerang effect.

FIELD NOTES

- Outdoors on a calm day, two or three loops from one hard throw is achievable.

TRIM: Doesn't close the loop? More bend or more power. Loops then crashes? Slightly less bend.

The Ring Wing

No wings, no fuselage — a flying hoop that shouldn't work.

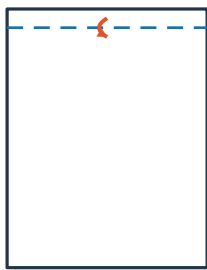
A rolled paper ring with a weighted rim, thrown like a football. The hoop's leading edge generates lift around its entire circumference while the rim weight keeps it gyroscopically stable. It looks impossible and flies beautifully — gentle, straight and slow.

SPEED	■ ■ ■ ■ ■
DISTANCE	■ ■ ■ ■ ■
AIRTIME	■ ■ ■ ■ ■
STUNTS	■ ■ ■ ■ ■

FOLD TIME ~3 MIN

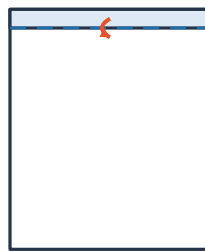
PAPER: Standard paper. (Half a sheet, cut or torn lengthwise, makes a tighter sportier ring.)

1



fold the top edge down about 2 cm

2



fold the band down twice more: a stiff rim

3



curl into a cylinder; tuck one rim end inside the other

With the paper in portrait, fold the top edge down about 2 cm.

Fold that band down twice more — a stiff weighted bar along the top edge. This becomes the ring's rim.

Curl the sheet into a cylinder, weighted bar at one end on the outside, and tuck one end of the bar inside the other end's folds. Work the seam until the hoop is round.

4



roll on a table edge until evenly round

5



a flying ring

spiral throw, like a tiny football

Shape it: roll the hoop gently on a table edge until it's an even circle with no kinks. Roundness = flight quality.

Grip it like a tiny football, rim forward, and throw with a smooth spiral spin. The spin is the stability — no spin, no flight.

The Tumblewing

A walkalong glider you can surf on a sheet of cardboard.

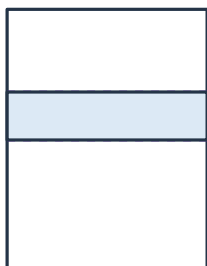
The strangest aircraft here: a strip of paper that tumbles end-over-end, generating lift as it rotates. Drop it and it falls slowly forward — but walk behind it holding a big piece of cardboard and your moving wall of air will keep it aloft indefinitely. This is 'walkalong gliding,' and it feels like levitating a paper bird.

SPEED	
DISTANCE	
AIRTIME	
STUNTS	

FOLD TIME ~2 MIN

PAPER: The lightest paper you can find — phonebook pages and tissue-weight gift paper are ideal. Cut a strip about 5 x 20 cm.

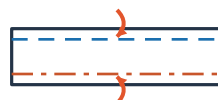
1



cut a light strip, about 5 x 20 cm

Cut a strip of very light paper, about 5 cm tall and 20 cm long (2 x 8 in).

2



bend long edges: one up, one down

Bend (don't crease hard) the long top edge up about 30° and the long bottom edge down about 30° — a shallow Z in cross-section.

3



tip both short ends up a few degrees

Tip both short ends up a few degrees, like a shallow tray.

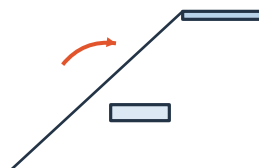
4



release with the gentlest backward roll

Launch: hold it high, top edge tilted slightly toward you, and release with the gentlest backward roll. It should tumble toward you... wait, away from you — rotating backward over its long axis as it drifts forward and down.

5



walk forward: the board's air wave carries it - surfing!

The magic part: hold a large sheet of cardboard at an angle behind and below the glider as you walk forward. The deflected air rises over the board and carries the Tumblewing — steer by moving the board. With practice you can fly laps of a room without it ever touching down.

The Canard

A fighter-jet layout with little wings up front.

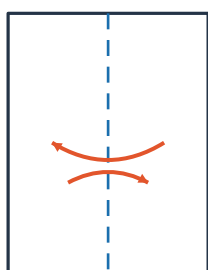
Real canard aircraft — think Eurofighter or Saab Viggen — put small control surfaces ahead of the main wing. This paper version does the same: nose fins folded out front, big delta behind. It looks sensational and flies with a fast, flat, darting character all its own.

SPEED ■■■■ ■
 DISTANCE ■■■ ■■
 AIRTIME ■■■ ■■ ■
 STUNTS ■■■ ■■ ■■

FOLD TIME ~7 MIN

PAPER: Standard or light paper;
 the nose has many layers.

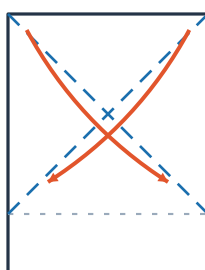
1



fold in half lengthwise, then unfold

Fold in half lengthwise and unfold.

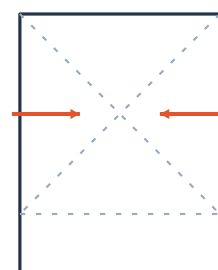
2



crease both diagonals of the top square

Fold and unfold both diagonals in the top square.

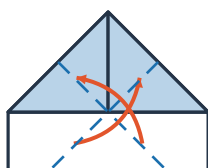
3



push sides in; top collapses into a triangle

Collapse the top into a layered triangle (waterbomb base): push the sides in along the creases and flatten.

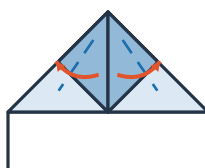
4



fold the front corners up to the peak

Fold the front-layer corners of the triangle up to the peak — a diamond sits on the triangle.

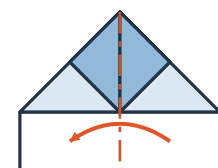
5



swing the small flaps back OUTWARD: fins

Now fold those small flaps back OUTWARD at an angle, so two little fins point forward-and-out from the nose. These are your canards.

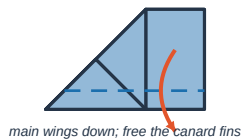
6



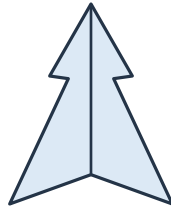
fold in half; fins stick out each side

Fold the plane in half away from you, letting the canard fins stick out on each side.

7



8



Fold the main wings down leaving ~2 cm of fuselage. Free the canard fins so they sit level, just ahead of and below the wing line.

Level the main wings, square the canards, and throw fast and flat. Expect quick, flat, fighter-like lines — and adjust the canards for pitch the way other planes use elevators.

FIELD NOTES

- Tilting both canards up a touch gives a dramatic swooping climb-out.
- The coolest-looking plane in the hangar on a shelf, full stop.

TRIM: Pitch is canard-controlled: nose drops = canards up a hair; stalls = canards down.

The Hurricane

A swept flying wing with no body at all.

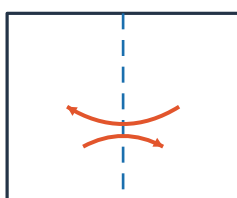
No fuselage, no nose point — the Hurricane is a pure flying wing, swept like a stealth bomber. Folded from a sheet turned sideways, it rides on a wide cushion of lift and lands so softly it can settle onto a desk without disturbing the papers.

SPEED	■ ■ ■ ■ ■
DISTANCE	■ ■ ■ ■ ■
AIRTIME	■ ■ ■ ■ ■
STUNTS	■ ■ ■ ■ ■

FOLD TIME ~5 MIN

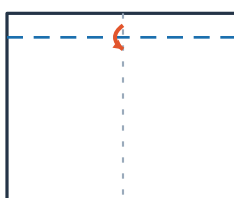
PAPER: Standard paper, landscape orientation.

1



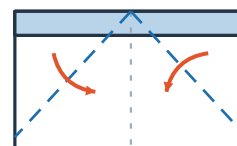
landscape: crease the center, unfold

2



fold the top edge down 2.5 cm, twice

3



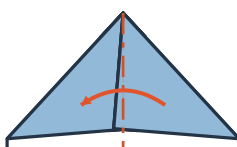
sweep each side down and back

Turn the sheet sideways (landscape). Fold in half left-to-right and unfold.

Fold the long top edge down about 2.5 cm, crease hard, and fold it down once more — a stiff leading-edge spar.

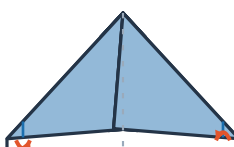
Fold each half down and back along a diagonal from the top center, sweeping the wing like a delta. The leading-edge spar now runs along both swept edges.

4



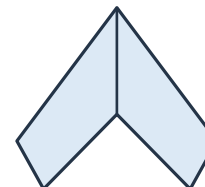
soft mountain crease down the center

5



fold small fins up at each wingtip

6



no throwing: a gentle level push

Add a soft mountain crease down the center line so the wing holds a shallow upside-down 'V' from the front.

Fold small vertical fins up at each wingtip for tracking.

Hold it at the center crease from above, level, and push it forward gently. No hard throws — the Hurricane is all glide.

FIELD NOTES

- Brilliant from a top bunk or balcony.
- Its slow landings make it the best plane for 'land on the target' games.

TRIM: Tuck the trailing edge center up slightly if it dives; flying wings are pitch-sensitive, so adjust gently.

The Dragonfly

Slim dart body, double wing surfaces — agile and odd.

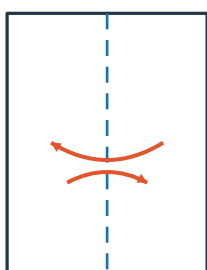
The Dragonfly pairs a slim dart fuselage with stacked wing surfaces: main wings plus upturned secondary fins that flex in flight. The result darts and flits with quick direction changes off the smallest trim adjustments — twitchy, insectile, and great fun in a large room.

SPEED	■ ■ ■ ■ ■
DISTANCE	■ ■ ■ ■ ■
AIRTIME	■ ■ ■ ■ ■
STUNTS	■ ■ ■ ■ ■

FOLD TIME ~6 MIN

PAPER: Light paper keeps the fins lively.

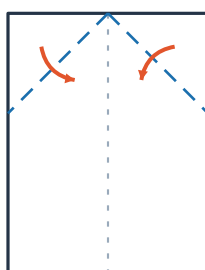
1



fold in half lengthwise, then unfold

Fold in half lengthwise and unfold.

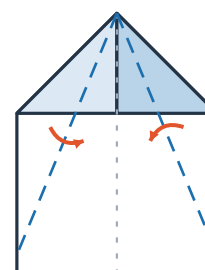
2



fold the top corners to the center crease

Fold the top corners to the center crease.

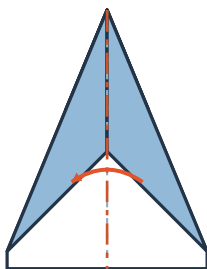
3



fold the angled edges to the center again

Fold the angled edges in to the center once more.

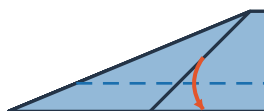
4



fold the plane in half away from you

Fold in half away from you.

5



HIGH crease: a deep 3 cm fuselage

Fold the wings down with a HIGH crease, leaving a deep 3 cm fuselage — deeper than any other plane here.

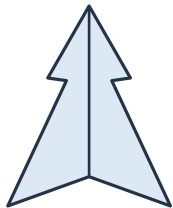
6



fold the outer HALF of each wing back up

Now fold the outer half of each wing back UP, creating a second surface that angles skyward — the dragonfly's double wings.

7

*quick, flat flick*

Level the inner wings, let the outer fins angle up symmetrically, and launch with a quick flat flick. Trim the fins to change its personality between flights.

FIELD NOTES

- Angle both fins far up for slow stable flight; flatten them for speed.
 - Mismatched fins make it juke unpredictably — fun on purpose.
- TRIM:** This plane is its fins: every adjustment of their angle is a new aircraft.



Hangar doors open.

25 designs, one sheet at a time. Fly them all, then start improving them — every great design in this book began as somebody's tweak to a dart.